test-588

October 11, 2026

[4]:
```
prec = 3400
CC = ComplexBallField(prec)
RBF = RealBallField(prec)
Pi = CC(pi, 0)
load('cubic_field_K_disc_-588.sage')
load('BCG_cubic_git.sage')
```

```
Loaded 'cubic_field_K_disc_-588.sage'. Contains the following variables:

K: Number Field in y with defining polynomial y^3 - y^2 + 5*y + 1
K has discriminant  -588 = -1 * 2^2 * 3 * 7^2
K has class number  3

pol_K = defining polynomial =  y^3 - y^2 + 5*y + 1
K_basis_SAGE = basis of K =  [1, y, y^2]

yRR = unique real root of pol_K  =  -0.191282440060928 …
yCC = unique complex root of pol_K with positive imaginary part
    =  0.595641220030464 … + I *  2.20750604731611 …
sigmaC = complex embedding of K sending y to yCC
K_basis_SAGE_sigmaC = [1, yCC, yCC^2]
K_basis_SAGE_sigmaR = [1, yRR, yRR^2]

ff =  Fractional ideal (3, y + 1)  of norm  3
q_ff = smallest positive integer in ff = 3
The narrow ray class group of conductor ff has order 6.

aa =  Fractional ideal (5, y - 2)
N_aa = norm of aa =  5
aa_comp = N_aa * aa^(-1)
bb =  Fractional ideal (2, y + 1)  of norm  2
(bb generates the narrow ray class group modulo ff.)

vareps = generator of totally positive units in 1+ff =  -y
varepsRR = real image of vareps =  0.191282440060928
varepsCC = image of vareps under sigmaC
or_Matrix = Matrix with entries the real and complex images of K_basis_SAGE
Eps =  matrix of the fundamental unit vareps in the basis K_basis_SAGE =
```

```
[ 0 -1  0]
[ 0  0 -1]
[ 1  5 -1]

N(aa) = 5 is an odd prime, as (2.9) asks.
aa is coprime to ff, as (2.9) asks.
sigmaR(vareps) = 0.19128 lies in (0,1), as section 2.2 asks.
vareps lies in 1 + ff, as section 2.2 asks.
q_ff = 3 is a positive integer of ff, as section 2.2 asks.
```

[6]:
```
%%time
L = ff
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
11.307     2     0.17687     13019        (-6, -3, -3)
11.307     2     0.17687     13019        (-27, 9, -6)
11.307     2     0.17687     13019        (-3, -9, 6)
11.307     2     0.17687     13019        (-144, 33, -27)
16.961     1     0.058958    39055        (21, 3, 3)
CPU times: user 1.58 s, sys: 39.5 ms, total: 1.61 s
Wall time: 1.61 s
```

[7]:
```
%%time
N_Ell_Gamma = 14000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (-6, -3, -3)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (-2, -1, -1)  with adm = (-6, -3, -3)
The wedge for  (-2, -1, -1)  is
a =  (-1, -1, 3)
b =  (-1, 1, 1)
alpha = (-7, -4, -3) , alpha(b) =  0 alpha(a) =  2
beta = (1, 2, 1) , beta(a) =  0 beta(b) =  2
gamma = (-6, -3, -3)
(alpha,beta,gamma) is positively oriented basis
tau =  1.0936239187 - 0.14716706982*I
```

```
sigma =  -0.47426238040 + 0.028150476212*I
The set F has  2 elements. They are
[-3 -1 -1]
[ 0  0  0]
(z, tau, sigma) in K =  [1/30*y^2 - 1/15*y + 5/6, -1/15*y + 17/15, 1/15*y^2 -
1/15*y - 2/15]
(z, tau, sigma) in K =  [1/3, -1/15*y + 17/15, 1/15*y^2 - 1/15*y - 2/15]
Wedge positively oriented with respect to h
certified decimals =  1017
u= -0.22978113472656073 - 0.33728122868081218*I
1/u= -1.3795863226741623 + 0.0250077123890153*I
P= x^18 + 12*x^17 + 106*x^16 + 219*x^15 + 284*x^14 + 16*x^13 + 1208*x^12 +
3414*x^11 + 2314*x^10 + 477*x^9 + 2314*x^8 + 3414*x^7 + 1208*x^6 + 16*x^5 +
284*x^4 + 219*x^3 + 106*x^2 + 12*x + 1

PK= x^6 + (3*y + 3)*x^5 + (-y^2 + y + 2)*x^4 + (7*y^2 + 3*y + 1)*x^3 + (-y^2 + y
+ 2)*x^2 + (3*y + 3)*x + 1
CPU times: user 10.9 s, sys: 35  s, total: 10.9 s
Wall time: 10.9 s
```

[8]:
```
%%time
L = ff*bb^(-1)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
11.307     1     0.088437    26037        (-3, -9, 6)
11.307     1     0.088437    26037        (-6, -3, -3)
11.307     1     0.088437    26037        (-27, 9, -6)
16.961     3     0.17687     13019        (-9/2, -6, 3/2)
16.961     3     0.17687     13019        (-33/2, 3, -9/2)
CPU times: user 1.79 s, sys: 6.75 ms, total: 1.8 s
Wall time: 1.8 s
```

[9]:
```
%%time
N_Ell_Gamma = 27000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (-3, -9, 6)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
```

```
Evaluating at h = adm/3 = (-1, -3, 2)  with adm = (-3, -9, 6)
The wedge for  (-1, -3, 2)  is
a =  (-1, 1, 1)
b =  (-5, 1, -1)
alpha = (-9/2, -13, 19/2) , alpha(b) =  0 alpha(a) =  1
beta = (1/2, 2, -3/2) , beta(a) =  0 beta(b) =  1
gamma = (-3, -9, 6)
(alpha,beta,gamma) is positively oriented basis
tau =  1.5468119593 - 0.073583534911*I
sigma =  -0.23713119020 + 0.014075238106*I
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [1/3, -1/30*y + 47/30, 1/30*y^2 - 1/30*y - 1/15]
Wedge positively oriented with respect to h
certified decimals =  1018
u= 0.39512989813388155 + 0.37636689581574047*I
1/u= 1.3269203008036179 - 1.2639106201959234*I
P= x^18 + 12*x^17 + 106*x^16 + 219*x^15 + 284*x^14 + 16*x^13 + 1208*x^12 +
3414*x^11 + 2314*x^10 + 477*x^9 + 2314*x^8 + 3414*x^7 + 1208*x^6 + 16*x^5 +
284*x^4 + 219*x^3 + 106*x^2 + 12*x + 1

PK= x^6 + (3*y + 3)*x^5 + (-y^2 + y + 2)*x^4 + (7*y^2 + 3*y + 1)*x^3 + (-y^2 + y
+ 2)*x^2 + (3*y + 3)*x + 1
CPU times: user 11.3 s, sys: 1.19 ms, total: 11.3 s
Wall time: 11.3 s
```

[10]:
```
%%time
L = ff*bb^(-2)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
5.6537     1     0.17687     13019        (3, 3/2, 3/2)
5.6537     1     0.17687     13019        (3/2, 9/2, -3)
5.6537     1     0.17687     13019        (27/2, -9/2, 3)
5.6537     1     0.17687     13019        (72, -33/2, 27/2)
33.922     1     0.029479    78110        (21, 3, 3)
CPU times: user 1.56 s, sys: 31.6 ms, total: 1.59 s
Wall time: 1.59 s
```

[11]:
```
%%time
N_Ell_Gamma = 13020
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
```

```
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (3, 3/2, 3/2)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (1, 1/2, 1/2)  with adm = (3, 3/2, 3/2)
The wedge for  (1, 1/2, 1/2)  is
a =  (-1, -1, 3)
b =  (-1, 1, 1)
alpha = (-1/2, -1/2, 0) , alpha(b) =  0 alpha(a) =  1
beta = (1/2, 1, 1/2) , beta(a) =  0 beta(b) =  1
gamma = (-3, -3/2, -3/2)
(alpha,beta,gamma) is positively oriented basis
tau =  0.093623918665 - 0.14716706982*I
sigma =  -0.47426238040 + 0.028150476212*I
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [-1/3, -1/15*y + 2/15, 1/15*y^2 - 1/15*y - 2/15]
Wedge negatively oriented with respect to h
certified decimals =  1001
u= -0.060550194201394607 + 0.094055752954422871*I
1/u= -4.8390562074267738 - 7.5167566542307835*I
P= x^18 + 12*x^17 + 106*x^16 + 219*x^15 + 284*x^14 + 16*x^13 + 1208*x^12 +
3414*x^11 + 2314*x^10 + 477*x^9 + 2314*x^8 + 3414*x^7 + 1208*x^6 + 16*x^5 +
284*x^4 + 219*x^3 + 106*x^2 + 12*x + 1

PK= x^6 + (3*y + 3)*x^5 + (-y^2 + y + 2)*x^4 + (7*y^2 + 3*y + 1)*x^3 + (-y^2 + y
+ 2)*x^2 + (3*y + 3)*x + 1
CPU times: user 5.47 s, sys: 3.93 ms, total: 5.47 s
Wall time: 5.47 s
```

[12]:
```
%%time
L = ff*bb^(-3)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
11.307     2     0.17687     13019        (3, 3/2, 3/2)
11.307     2     0.17687     13019        (27/2, -9/2, 3)
11.307     2     0.17687     13019        (3/2, 9/2, -3)
11.307     2     0.17687     13019        (72, -33/2, 27/2)
16.961     1     0.058958    39055        (-21/2, -3/2, -3/2)
CPU times: user 1.52 s, sys: 40 ms, total: 1.56 s
Wall time: 1.56 s
```

[13]:
```
%%time
N_Ell_Gamma = 13020
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (3, 3/2, 3/2)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (1, 1/2, 1/2)  with adm = (3, 3/2, 3/2)
The wedge for  (1, 1/2, 1/2)  is
a =  (-2, -2, 6)
b =  (-2, 2, 2)
alpha = (-7/2, -2, -3/2) , alpha(b) =  0 alpha(a) =  2
beta = (1/2, 1, 1/2) , beta(a) =  0 beta(b) =  2
gamma = (-3, -3/2, -3/2)
(alpha,beta,gamma) is positively oriented basis
tau =  1.0936239187 - 0.14716706982*I
sigma =  -0.47426238040 + 0.028150476212*I
The set F has  2 elements. They are
[-3/2 -1/2 -1/2]
[   0    0    0]
(z, tau, sigma) in K =  [1/30*y^2 - 1/15*y + 1/6, -1/15*y + 17/15, 1/15*y^2 -
1/15*y - 2/15]
(z, tau, sigma) in K =  [-1/3, -1/15*y + 17/15, 1/15*y^2 - 1/15*y - 2/15]
Wedge negatively oriented with respect to h
certified decimals =  1001
u= -1.3795863226741623 + 2.0250077123890153*I
1/u= -0.22978113472656073 - 0.33728122868081218*I
P= x^18 + 12*x^17 + 106*x^16 + 219*x^15 + 284*x^14 + 16*x^13 + 1208*x^12 +
3414*x^11 + 2314*x^10 + 477*x^9 + 2314*x^8 + 3414*x^7 + 1208*x^6 + 16*x^5 +
284*x^4 + 219*x^3 + 106*x^2 + 12*x + 1

PK= x^6 + (3*y + 3)*x^5 + (-y^2 + y + 2)*x^4 + (7*y^2 + 3*y + 1)*x^3 + (-y^2 + y
+ 2)*x^2 + (3*y + 3)*x + 1
CPU times: user 9.95 s, sys: 1.23 ms, total: 9.95 s
Wall time: 9.95 s
```

[14]:
```
%%time
L = ff*bb^(-4)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
```

```
decimals:
s/rho      s     rho          N_Ell_Gamma  adm
11.307     1     0.088437     26037        (3/2, 9/2, -3)
11.307     1     0.088437     26037        (3, 3/2, 3/2)
11.307     1     0.088437     26037        (27/2, -9/2, 3)
16.961     3     0.17687      13019        (9/4, 3, -3/4)
16.961     3     0.17687      13019        (33/4, -3/2, 9/4)
CPU times: user 1.57 s, sys: 13.6 ms, total: 1.58 s
Wall time: 1.58 s
```

[15]:
```
%%time
N_Ell_Gamma = 26040
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (3/2, 9/2, -3)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (1/2, 3/2, -1)  with adm = (3/2, 9/2, -3)
The wedge for  (1/2, 3/2, -1)  is
a =  (-2, 2, 2)
b =  (-10, 2, -2)
alpha = (-9/4, -13/2, 19/4) , alpha(b) =  0 alpha(a) =  1
beta = (1/4, 1, -3/4) , beta(a) =  0 beta(b) =  1
gamma = (-3/2, -9/2, 3)
(alpha,beta,gamma) is positively oriented basis
tau =  1.5468119593 - 0.073583534911*I
sigma =  -0.23713119020 + 0.014075238106*I
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [-1/3, -1/30*y + 47/30, 1/30*y^2 - 1/30*y - 1/15]
Wedge negatively oriented with respect to h
certified decimals =  1001
u= 1.3269203008036179 - 1.2639106201959234*I
1/u= 0.39512989813388155 + 0.37636689581574047*I
P= x^18 + 12*x^17 + 106*x^16 + 219*x^15 + 284*x^14 + 16*x^13 + 1208*x^12 +
3414*x^11 + 2314*x^10 + 477*x^9 + 2314*x^8 + 3414*x^7 + 1208*x^6 + 16*x^5 +
284*x^4 + 219*x^3 + 106*x^2 + 12*x + 1

PK= x^6 + (3*y + 3)*x^5 + (-y^2 + y + 2)*x^4 + (7*y^2 + 3*y + 1)*x^3 + (-y^2 + y
+ 2)*x^2 + (3*y + 3)*x + 1
CPU times: user 10.9 s, sys: 6  s, total: 10.9 s
```

```
Wall time: 10.9 s
```

[16]:
```
%%time
L = ff*bb^(-5)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
5.6537     1     0.17687     13019        (-3/2, -3/4, -3/4)
5.6537     1     0.17687     13019        (-3/4, -9/4, 3/2)
5.6537     1     0.17687     13019        (-27/4, 9/4, -3/2)
5.6537     1     0.17687     13019        (-36, 33/4, -27/4)
33.922     1     0.029479    78110        (-21/2, -3/2, -3/2)
CPU times: user 1.49 s, sys: 32.1 ms, total: 1.52 s
Wall time: 1.52 s
```

[17]:
```
%%time
N_Ell_Gamma = 13020
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (-3/2, -3/4, -3/4)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (-1/2, -1/4, -1/4)  with adm = (-3/2, -3/4, -3/4)
The wedge for  (-1/2, -1/4, -1/4)  is
a =  (-2, -2, 6)
b =  (-2, 2, 2)
alpha = (-1/4, -1/4, 0) , alpha(b) =  0 alpha(a) =  1
beta = (1/4, 1/2, 1/4) , beta(a) =  0 beta(b) =  1
gamma = (-3/2, -3/4, -3/4)
(alpha,beta,gamma) is positively oriented basis
tau =  0.093623918665 - 0.14716706982*I
sigma =  -0.47426238040 + 0.028150476212*I
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [1/3, -1/15*y + 2/15, 1/15*y^2 - 1/15*y - 2/15]
Wedge positively oriented with respect to h
certified decimals =  1001
u= -4.8390562074267738 - 7.5167566542307835*I
1/u= -0.060550194201394607 + 0.094055752954422871*I
P= x^18 + 12*x^17 + 106*x^16 + 219*x^15 + 284*x^14 + 16*x^13 + 1208*x^12 +
```

```
3414*x^11 + 2314*x^10 + 477*x^9 + 2314*x^8 + 3414*x^7 + 1208*x^6 + 16*x^5 +
284*x^4 + 219*x^3 + 106*x^2 + 12*x + 1

PK= x^6 + (3*y + 3)*x^5 + (-y^2 + y + 2)*x^4 + (7*y^2 + 3*y + 1)*x^3 + (-y^2 + y
+ 2)*x^2 + (3*y + 3)*x + 1
CPU times: user 5.47 s, sys: 3.98 ms, total: 5.47 s
Wall time: 5.47 s
```

[ ]: